

0.1 Dual Space

0.1.1 Definition

Definition 0.1.1. Let V be a Vector Space over a field F .
A Vector Space

$$\mathcal{L}(V, F) \stackrel{\text{def}}{=} \{T : V \rightarrow F \mid T \text{ is Linear Map}\}$$

is called *Dual Space* of V , denoted as V^* .

Note: Naturally, we can consider the Dual of Dual, $(V^*)^*$, it is denoted as V^{**} .

Definition 0.1.2. Let $\beta \subset V$ be a basis for V . Denote $\beta = \{x_i \mid i \in I\}$.
For each $i \in I$, define *Coordinate function*:

$$\mathbf{f}_i : V \rightarrow F :$$